



WORKSHEET 3F

QUESTION 1

Write a count-controlled loop to display a row of 80 stars.

```
MOV CX, 80
```

```
PRINT_STAR:
    MOV DL, '*'
    MOV AH, 02h
    INT 21h
    LOOP PRINT_STAR
```

QUESTION 2

What happens if CX is 0 when the a For loop is entered

If CX = 0 before executing LOOP:

- LOOP first decrements CX
- CX becomes FFFFh
- Since CX $\neq$ 0, the loop executes 65,536 times

QUESTION 3

Write some code to count the number of characters in an input line. Use a while loop.

```
MOV CX, 0 ; character count
```

```
READ_CHAR:
    MOV AH, 01h
    INT 21h ; read character
    CMP AL, 0Dh ; Enter key?
    JE DONE
    INC CX
    JMP READ_CHAR
```

```
DONE:
```



QUESTION 4

Use ROL to count the number of 1 bits in BX, without changing BX. Put the answer in AX.

```
MOV AX, 0           ; bit count
MOV CX, 16          ; 16 bits
MOV DX, BX          ; copy BX (original preserved)
```

```
COUNT_BITS:
    ROL DX, 1
    JNC SKIP
    INC AX
SKIP:
    LOOP COUNT_BITS
```

QUESTION 5

Suppose DH contains 8Ah, CF= 1, and CL contains 3. What are the values of DH and CF after the instruction RCR DH, CL is executed?

- DH = 71h = 01110001
- CF = 0

QUESTION 6

If AL contains 11011100, we want to make it 00111011 (i.e., reverse the pattern). Write the assembly code to do so.

```
MOV BL, AL
MOV AL, 0
MOV CX, 8

REVERSE:
    RCR BL, 1
    RCL AL, 1
    LOOP REVERSE
```



QUESTION 7

- Binary Input: write a program to read in an 8-bit binary number.
- Conversion in bit value needs to be done as soon as the input character is entered.
- After that collect the bits in register.

```
MOV BL, 0
MOV CX, 8

READ_BIT:
    MOV AH, 01h
    INT 21h          ; read '0' or '1'

    SUB AL, '0'      ; ASCII → binary
    SHL BL, 1
    OR BL, AL

    LOOP READ_BIT
```

QUESTION 8

- Binary OUTPUT: Write a program to print out the contents of BL bit by bit.

```
MOV CX, 8

PRINT_BITS:
    ROL BL, 1
    JC PRINT_ONE
    MOV DL, '0'
    JMP PRINT

PRINT_ONE:
    MOV DL, '1'

PRINT:
    MOV AH, 02h
    INT 21h
    LOOP PRINT_BITS
```